

Work Plan: Replicating the Gini-Coefficient Analysis of Richardson et al. (2024)

Goal. Reproduce the core statistical result of Richardson, Kahn & Kalmus (2024, *Comms Earth & Environ* 5:472): the Gini coefficient quantifying how well AIRS-FCST CAPE predicts heavy hourly precipitation (MRMS QPE exceedances) over the central-eastern CONUS — specifically the analyses behind their Fig. 2 (Gini derivation and percentile sweep) and Fig. 3a (Gini vs. forecast hour, with significance).

Design constraint (non-negotiable). All code must be **maximally interpretable** — it is the foundation for extending this analysis to SMAP soil-moisture predictors. Concretely:

1. **Predictor-agnostic core.** The Gini machinery must accept *any* 1-D predictor array + event flags, never "CAPE" specifically. Swapping in `SMAP_smsfc_av` must require zero changes to the statistics code.
 2. **One function = one paper sentence.** Each step of the paper's Methods maps to one small, pure function whose docstring quotes/cites the exact passage it implements (e.g. "*Methods* → '*Gini and significance calculations*', ¶1").
 3. **All constants in one config file**, each with a comment citing its source in the paper (domain bounds, QC rules, percentile lists, bin count, bootstrap reps).
 4. **No hidden state.** The data layer produces one tidy analysis table (rows = day × hour × grid cell; columns = predictors + QPE + masks); everything downstream consumes that table.
 5. **Unit tests with analytically known answers** for the Gini function (uniform → 0, perfectly sorted → →1, half-sorted → known value).
 6. **Notebooks are thin.** Each notebook reproduces one figure/claim by calling library functions; no analysis logic lives in notebooks.
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1. The method being replicated (distilled from the paper)

Paper framing: AIRS temperature/humidity profiles, advected forward with HYSPLIT + WRF27km winds ("trajectory enhancement"), yield hourly $1^\circ \times 1^\circ$ CAPE fields (AIRS-FCST). Skill is measured as the nonuniformity of $P(\text{conv} \mid q, T)$ via the Gini coefficient, with MRMS gauge-corrected hourly QPE exceedances as the convection proxy.

Gini recipe (Methods → "Gini and significance calculations"):

1. Choose a QPE percentile threshold X (they sweep $X = 95 \rightarrow 99.95$; $\text{QPE}_{99.95} \approx 5.1 \text{ mm h}^{-1}$ in their sample). Thresholds are computed from the **entire pooled sample** — all locations, seasons, hours, wet and dry.
2. Flag each sample 1 if $\text{QPE} > \text{QPE}_X$, else 0.
3. Sort flags by the predictor value. Zero-valued predictors get a tiny random perturbation ($\sim \pm 1 \times 10^{-10}$) so sorting is unambiguous (this makes the CDF linear over the zero-CAPE range).
4. Compute the normalized cumulative distribution of events across **100 equal-count predictor bins** (each bin = 1% of samples).
5. $\text{Gini} = 2 \times \text{area between that event-capture curve and the 1:1 diagonal}$. (Equivalently $\text{Gini} = 2 \cdot \text{AUC} - 1$ of the ROC; the paper's refs 76–77 give the relationship. Implement the binned-CDF version to match the paper exactly; verify against the ROC identity as a cross-check.)

- Gini = 0 → uninformative predictor; Gini → 1 → all events at highest predictor values.

Significance (for the Fig. 3 hour-by-hour claims):

- *Bootstrap SE*: pool the predictor samples from all six forecast hours; resample with replacement down to a one-hour sample size; compute Gini; repeat 500×; SD of the 500 Ginis = 1σ . Hours treated as independent → SE of an hour-to-hour difference = $\sqrt{2}\cdot\sigma$; $p < 0.05$ threshold $\approx 2\sqrt{2}\cdot\sigma$ ($\approx \pm 0.08$ in their sample).
- *Trend test*: OLS slope of Gini vs. forecast hour within a product; $|\text{slope}| > 2\times\text{SE}(\text{slope}) \rightarrow$ significant at $p < 0.05$.
- Expected qualitative result: AIRS-FCST Gini **improves significantly with forecast hour**; overpass-time AIRS CAPE degrades.

Sample-selection rules (Methods → "Calculation of thermodynamic indices and selection of data for analysis"):

- Predictor = **MU_CAPE only** (their sensitivity tests found it most consistently predictive; MML/SFC parcels and CIN/EL/LCL/LFC added ~no skill).
- Analysis hours = the **six forecast hours 21–02 UTC** (not the overpass snapshot).
- Grid cells kept for a day only if valid at **all** timesteps (so the geographic sample is identical for every forecast hour): (i) > 20 AIRS-FCST parcels in the profile, and (ii) valid MU_CAPE, MU_EL, MU_LCL, MU_CIN (zero counts as valid).
- Domain: **32–53°N, 107–64°W** (data south of 32°N excluded for poor performance), **land fraction $\geq 50\%$** .
- Season: March–November. Paper years: **2019–2020**. Their pooled N > 1 million (> 160k per forecast hour).

2. Mapping the paper onto our data

Local data (DVC-tracked): `data/FCST_SMAP_MRMS_{2016..2021}.nc` (six years) + `data/lsm.nc`.

Each yearly file: dims (`date`: 366, `nhours`: 7, `lat`: 28, `lon`: 43); grid 25.5–52.5°N × 106.5–64.5°W at 1°; valid days $\approx$ day-of-year 60–334 (March–November).

Paper quantity	Local variable	Notes
AIRS-FCST MU CAPE (predictor)	<code>FCST_MU_CAPE</code>	J/kg. <code>FCST_MML_*</code> also present for the sensitivity check.
QC companions	<code>FCST_MU_CIN</code> , <code>FCST_MU_EL</code> , <code>FCST_MU_LCL</code>	required-valid per paper QC
Parcel count per profile	<code>FCST_N</code>	Δ appears to be all zeros in 2016 where CAPE is valid — see Open Question Q2
MRMS hourly QPE (target)	<code>MRMS_GaugeCorrQPE01H_av</code>	1° grid-cell mean , matching the paper's "1° × 1° grid-cell mean QPE"
Forecast hours 21–02 UTC	<code>nhours</code> slots 1–6	verified via <code>FCST_parceltime</code> : slot 0 = AIRS overpass (~17:45 UTC snapshot $\Rightarrow$ the paper's

Paper quantity	Local variable	Notes
		"AIRS" proximity-sounding baseline), slots 1–6 = 21,22,23,00,01,02 UTC
Land fraction ≥ 50%	<code>data/lsm.nc</code> (<code>lsm</code> , 1° global)	subset to our grid; threshold at 0.5
32–53°N restriction	slice <code>lat ≥ 32</code>	our grid starts at 25.5°N, so this cut must be applied
SMAP extension fields	<code>SMAP_*</code> (overpass-relative hours) and <code>SMAP_L4_*</code> (fixed hours 16.5–28.5 UTC)	soil moisture av/sd, W-E & S-N & abs gradients, layer-1 T/q/u, precip flux — the predictor menu for Phase 6
Precip-type context	<code>MRMS_PrecipFlag_cnt</code>	12 flags incl. <code>Convection</code> , <code>Hail</code> — useful later for verifying "heavy QPE ≈ convection"

Not in our files: ERA5 CAPE/QPE, HRRR QPE, and a separate AIRS-overpass CAPE variable (though `nhours` slot 0 is overpass-time CAPE, replicated per the paper's method). So the full Fig. 2b/3 multi-product comparison is out of scope; the replication targets the **AIRS-FCST rows** of those figures. ERA5/HRRR comparators could be added later from the paper's JPL Open Repository dataset if desired.

Known differences from the paper (expect small numerical departures):

- We have **2016–2021** (~3× the paper's sample). For validation, first run on the **2019–2020 subset** to match the paper; then re-run on all six years as the "extended replication."
- September 2021 onward is affected by the AIRS deep-space-maneuver retrieval issue the paper cites — decide whether to truncate 2021 (recommend: run 2021 both ways and compare).
- Our files come from the FCST_SMAP_MRMS pipeline cut, which may embed slightly different upstream QC than the paper's exact dataset ("clean data" per project notes).

3. Open questions to resolve in Phase 1 (data audit)

- Q1 — hour-slot semantics:** confirm slot 0 = overpass for all years, and that slots 1–6 are always 21–02 UTC (check a sample of days per year via `FCST_parceltime`).
- Q2 — `FCST_N`:** in 2016 it is 0 wherever CAPE is valid, so the paper's ">20 parcels" screen can't be applied as-is. Determine what `FCST_N` encodes (ask data provider / check other years). Working assumption: the parcel-count QC was already applied upstream in this "clean" cut; document whichever conclusion holds.
- Q3 — threshold sanity check:** does our pooled 2019–2020 QPE_{99.95} land near the paper's 5.1 mm h⁻¹? A large discrepancy signals a units or sampling mismatch.
- Q4 — sample size:** paper reports >160k per forecast hour for 2019–2020 after QC; check ours matches to within ~10%.
- Q5 — 2016/2020 leap-day handling** (366-day files for non-leap years contain a padding day; confirm it is all-NaN and dropped by masking).

4. Repository layout & interpretability rules

```

JPL_AIRS/
├─ WORKPLAN.md
├─ environment.yml
from base env!)
├─ src/convection_skill/
throughout
|   └─ config.py
comment:
|   |
"restricted to north of 32N"
|   |
equally sized bins"
|   |
10, ...
|   └─ data_loading.py
|   |
lat, lon,
|   |
land_frac]
|   └─ quality_control.py
named function
|   └─ gini.py
|   |
bool[]
|   |
(x, cdf)
|   |
(convenience)
|   └─ significance.py
|   └─ plotting.py
├─ tests/
|   └─ test_gini.py
vs sklearn
|   └─ test_quality_control.py
├─ notebooks/ (or scripts/ – plain .py with # %% cells also fine)
|   └─ 01_data_audit.ipynb
threshold check
|   └─ 02_fig2_replication.ipynb
hours)
|   └─ 03_fig3_replication.ipynb
trend
|   └─ 04_smap_first_look.ipynb
└─ results/
large
# this file
# pinned conda env (netCDF4/xarray missing
# importable package, pure functions
# ALL constants, each with a paper citation
# DOMAIN_LAT = (32, 53) # Methods:
# N_BINS = 100 # Methods: "100
# PERCENTILES = [95, 99, 99.5, 99.9, 99.95]
# BOOTSTRAP_REPS = 500, TIEBREAK_EPS = 1e-
# open_year(), build_analysis_table()
# → one tidy DataFrame: [date, hour_utc,
# mu_cape, mu_cin, ..., qpe, smap_*,...,
# apply_paper_qc(table) – each rule its own
# PREDICTOR-AGNOSTIC core:
# exceedance_flags(values, percentile) ->
# detection_cdf(predictor, flags, n_bins) ->
# gini_from_cdf(x, cdf) -> float
# gini(predictor, flags) -> float
# bootstrap_gini_se(), hourly_trend_test()
# fig2a_style(), fig2b_style(), fig3_style()
# analytic cases; gini == 2*AUC-1 cross-check
# tiny synthetic table exercising each QC rule
# Phase 1: answers Q1-Q5, coverage maps,
# CDF + Gini percentile sweep (pooled forecast
# per-hour Gini + bootstrap significance +
# Phase 6: same machinery, SMAP predictors
# derived tables (CSV/parquet) – DVC-track if

```

Code style contract: NumPy/pandas only in the core (no framework magic); type hints + docstrings everywhere; docstrings state *which paper sentence* the function implements and *why* (e.g. why the $\pm 1e-10$ perturbation exists); no function longer than ~40 lines; module-level `if __name__ == "__main__"` demos on the core modules so each is runnable/inspectable standalone.

5. Phased plan

Phase 0 — Environment (½ day). Create `environment.yml` (python, numpy, pandas, xarray, netcdf4, matplotlib, scipy, pytest, jupyter). Note: the base anaconda env lacks a netCDF backend — `h5py` works but `netcdf4` is the interpretable choice. Verify `xr.open_dataset` on one yearly file. Commit env + skeleton package.

Phase 1 — Data audit (1 day). `01_data_audit.ipynb` Resolve Q1–Q5. Produce: per-year valid-sample counts, map of mean MU_CAPE and QPE, histogram of QPE with percentile thresholds marked, land-mask plot after the $\geq 50\%$ cut. **Gate: do not proceed until Q1–Q3 are resolved and documented in the notebook.**

Phase 2 — Core library + tests (1–2 days). Implement `gini.py`, `quality_control.py`, `data_loading.py`, `significance.py` per the layout above. Tests must pass, including the Gini-vs-ROC identity check and a shuffle test (shuffled predictor $\rightarrow$ Gini $\approx 0 \pm$ noise).

Phase 3 — Fig. 2 replication (1 day). `02_fig2_replication.ipynb`, 2019–2020 subset, forecast hours pooled.

- 2a: event-capture CDF for $QPE_{99.95}$ vs. MU_CAPE percentile, Gini annotated.
- 2b: Gini vs. threshold percentile ($95 \rightarrow 99.95$) for AIRS-FCST CAPE (and overpass-CAPE from slot 0 as the available comparator).
- 2c: CDFs for several QPE_X on one axes. **Acceptance:** curve shapes match the paper (linear segment over zero-CAPE range; $\sim 80\%$ of $QPE_{99.95}$ events captured above $CAPE_{90}$); Gini values within $\sim \pm 0.05$ of the paper's AIRS-FCST values (bootstrap SE ~ 0.01 , plus dataset-cut differences).

Phase 4 — Fig. 3a replication (1 day). `03_fig3_replication.ipynb`, 2019–2020 subset. Per-hour (21–02 UTC) CDFs and Ginis for $QPE_{99.95}$; bootstrap SEs; hour-difference significance at $2\sqrt{2}\sigma$; OLS trend test. Also run slot-0 overpass CAPE per hour (its Gini should *decline* with hour, mirroring the paper's Fig. 3b). **Acceptance:** significant *increase* of AIRS-FCST Gini with hour and significant *decline* for overpass CAPE, per the paper's stated result; $\sigma \approx 0.02$ – 0.04 scale so that ± 0.08 is the $p < 0.05$ difference threshold.

Phase 5 — Extended replication (½ day). Re-run Phases 3–4 on all of 2016–2021 (with and without post-Aug-2021). Report how Gini and its uncertainty tighten with $3\times$ sample. This is the new-science baseline the SMAP work builds on.

Phase 6 — SMAP extension hooks (1 day, gateway to the real project). `04_smap_first_look.ipynb`
Demonstrate the machinery is truly predictor-agnostic:

- Gini of $QPE_{99.9}$ using each `SMAP_*` field alone (expect low — soil moisture is a modulator, not a CAPE substitute).
- **Stratified Gini:** CAPE $\rightarrow$ QPE Gini computed within SMAP soil-moisture terciles (dry/mid/wet) and, per the lit review's regime-dependence warnings (Tuttle & Salvucci 2016; Guillod et al. 2015), separately for east vs. plains longitude bands. Requires one new helper: `stratified_gini(table, predictor, stratifier, n_strata)`.
- Document the two natural extension paths: (a) conditioning/stratification, (b) bivariate predictors (e.g., CAPE $\times$ soil-moisture-gradient score) — the latter needs only a column-combining step before the same `gini()` call.

6. Verification & bookkeeping

- Each figure notebook ends with a "Comparison to paper" cell stating target value, obtained value, and pass/fail against the acceptance criteria above.
- Derived thresholds (QPE_X in mm/h per sample definition) written to `results/thresholds.csv` so every downstream analysis uses identical, versioned cut-offs.
- Code in git; `results/` and any regenerated data in DVC, consistent with the existing repo setup.
- Random seeds fixed in `config.py` for the tie-break perturbation and bootstrap (document that results must be insensitive to the seed — the paper states the perturbation does not affect results; verify with two seeds once).

7. References

- Richardson, M. T., B. H. Kahn, P. M. Kalmus (2024). *Mesoscale air motion and thermodynamics predict heavy hourly U.S. precipitation*. *Comms Earth & Environ* 5:472. doi:10.1038/s43247-024-01614-1 (PDF in repo root; Methods pp. 6–7 are the implementation spec).
- `convective_initiation_lit_review.md` (repo root) — mechanism menu and regime-stratification rationale for Phase 6.